UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 011 (VoP) · Method of variation of parameters · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011V·9	Unit 2 deck 011 “Variation of Parameters method” (SESSION 8), slide 9 (rendered .pptx)
T011V 00:03:26	AV-summary transcript “Variation of Parameters method” (14.9 min, listed first), segment start
T011Vb 00:16:53	AV-summary transcript “Variation of Parameters method_Continued” (24.5 min, second), segment start
CW U2 P25	Unit 1 & 2 class-work sheet, Unit 2 problem 25
ESA Jul-23 3c	Past paper: UE22MA141A July 2023, question 3.c (Jan-24 s1/s2 = UE23MA141A Jan 2024 sets 1/2)

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A Formula sheet

1 The method: $y_p = uy_1 + vy_2$ with the Wronskian

E1.1 · Standard form and CF

ESA ★★★

$$y'' + p\,y' + q\,y = X, \qquad y_c = c_1 y_1 + c_2 y_2$$

GIVES The starting point: y_1, y_2 from the CF.

APPLIES Coefficient of y'' equal to 1.

NOTE Divide by the leading coefficient first (CW P27).

S011V·4 · S011V·6 · T011V 00:00:43 · T011V 00:06:06 · CW U2 P27

E1.2 · Wronskian

ESA ★★★

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0$$

GIVES The denominator of both integrals.

APPLIES y_1, y_2 linearly independent.

NOTE Order matters: $W(y_2, y_1) = -W(y_1, y_2)$, but the PI is unchanged (T011Vb 00:09:20).

S011V·5 · S011V·8 · T011V 00:02:35 · T011V 00:10:22 · ESA Jul-23 3c

E1.3 · The parameters u and v

ESA ★★★

$$u = -\int \frac{y_2\,X}{W}\,dx, \qquad v = \int \frac{y_1\,X}{W}\,dx$$

GIVES The two varying parameters.

APPLIES Every VoP problem.

NOTE Minus sign on u (with y_2); no constants of integration.

S011V·5 · T011V 00:03:26 · T011V 00:04:18 · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E1.4 · The particular integral and the general solution

ESA ★★★

$$y_p = u\,y_1 + v\,y_2, \qquad y = c_1 y_1 + c_2 y_2 + y_p$$

GIVES The PI and, with the CF, the general solution.

APPLIES After E1.3.

NOTE Terms of y_p that are multiples of y_1 or y_2 may be absorbed into c_1, c_2.

S011V·5 · S011V·10 · S011V·17 · T011V 00:14:11 · T011Vb 00:16:16

E1.5 · Why: the two conditions on u', v' ESA ★

$$u'y_1 + v'y_2 = 0, \qquad u'y_1' + v'y_2' = X$$
$$\Rightarrow\ u' = -\frac{y_2 X}{W}, \quad v' = \frac{y_1 X}{W}$$

GIVES E1.3 by Cramer's rule.

APPLIES Derivation; not on the slides.

NOTE The first condition is a free choice that keeps y_p' free of u', v'; the second is the equation itself.

ADDED FOR CLARITY S011V·5

2 Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation

E2.1 · Integrals of slides 011V·12–16 ESA ★★

$$\int \sec 2x\,dx = \tfrac{1}{2}\log(\sec 2x + \tan 2x), \qquad \int \frac{dx}{e^x(1+e^x)} = -e^{-x} + \log(1+e^{-x})$$
$$\int \frac{e^x}{1+e^x}dx = \log(1+e^x), \qquad \int xe^{-4x}dx = -\frac{e^{-4x}}{4}\left(x + \frac{1}{4}\right)$$

GIVES The four integrals the deck needs.

APPLIES $\tan 2x$, $\frac{2}{1+e^x}$, xe^{-x}.

NOTE $\frac{1}{e^x(1+e^x)} = \frac{1}{e^x} - \frac{1}{1+e^x}$ and $\frac{1}{1+e^x} = \frac{e^{-x}}{1+e^{-x}}$.

S011V·9 · S011V·12 · S011V·15 · S011V·16 · S011V·17 · T011Vb 00:05:21 · T011Vb 00:11:06 · T011Vb 00:12:55

E2.2 · Result: $(D^2+4)y = \tan 2x$ ESA ★★

$$y_p = -\frac{1}{4}\cos 2x\,\log(\sec 2x + \tan 2x)$$

GIVES The PI (slide 011V·13's line above the box; CW P28).

APPLIES $u = \frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$, $v = -\frac{\cos 2x}{4}$.

NOTE Slide 011V·13's box omits $\cos 2x$.

S011V·13 · CW U2 P28 · T011Vb 00:07:06

E2.3 · Result: $(D^2-1)y = \frac{2}{1+e^x}$ ESA ★★

$$y_p = e^x\log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$$

GIVES The PI (slide 011V·17).

APPLIES $u = -e^{-x} + \log(1+e^{-x})$, $v = -\log(1+e^x)$.

NOTE The -1 is not a CF term: keep it.

S011V·17 · T011Vb 00:15:48

3 Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$

E3.1 · Complex roots $\alpha \pm i\beta$ with $X = e^{\alpha x}g(x)$ ESA ★★★

$$y_1 = e^{\alpha x}\cos\beta x, \quad y_2 = e^{\alpha x}\sin\beta x, \quad W = \beta e^{2\alpha x}$$
$$u = -\frac{1}{\beta}\int g\sin\beta x\,dx, \qquad v = \frac{1}{\beta}\int g\cos\beta x\,dx$$

GIVES The integrals with the exponentials already cancelled.

APPLIES CW P23, P25, P26, P28; ESA Jan-24 s1 3a, s2 3a.

NOTE $\alpha = 0$ for $D^2 + a^2$ ($W = a$); $D^2 - 2D + 2$: $1 \pm i$; $D^2 + 2D + 2$: $-1 \pm i$.

ADDED FOR CLARITY S011V·11 · CW U2 P23 · CW U2 P25 · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E3.2 · Trigonometric integrals used ESA ★★★

$$\int \tan ax\,dx = \tfrac{1}{a}\log\sec ax, \qquad \int \cot x\,dx = \log\sin x, \qquad \int \sec x\,dx = \log(\sec x + \tan x)$$
$$\int \frac{\sin^2 x}{\cos x}dx = \log(\sec x + \tan x) - \sin x, \qquad \int \frac{\sin x}{\cos^3 x}dx = \frac{\sec^2 x}{2}$$

GIVES Every integral of Topic 3.

APPLIES After E3.1.

NOTE Strictly $\log|\cdot|$; $\int \frac{\sin x}{\cos^3 x}dx = \frac{\tan^2 x}{2}$ is also right (differs by $\frac{1}{2}$).

S011V·12 · CW U2 P23 · CW U2 P25 · CW U2 P26 · ESA Jan-24 s2 3a

E3.3 · Results: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$ ESA ★★★

$$\frac{\sec ax}{D^2 + a^2}: \ \frac{\cos ax}{a^2}\log\cos ax + \frac{x}{a}\sin ax, \qquad \frac{\csc x}{D^2 + 1}: \ -x\cos x + \sin x\log\sin x$$
$$\frac{e^x\tan x}{D^2 - 2D + 2}: \ -e^x\cos x\log(\sec x + \tan x), \qquad \frac{e^{-x}\sec^3 x}{D^2 + 2D + 2}: \ \tfrac{1}{2}e^{-x}\sin x\tan x$$

GIVES The four PIs (W3.1–W3.4).

APPLIES Asked by variation of parameters.

NOTE The last one may also be written $-\frac{e^{-x}\cos 2x}{2\cos x}$ (differs by a CF term).

CW U2 P23 · CW U2 P25 · CW U2 P26 · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

4 Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap

E4.1 · Repeated root m with $X = e^{mx}g(x)$ ESA ★★★

$$y_1 = e^{mx}, \quad y_2 = xe^{mx}, \quad W = e^{2mx}, \qquad u = -\int x\,g\,dx, \quad v = \int g\,dx$$

GIVES The integrals with the exponentials cancelled.

APPLIES $(D-m)^2y = e^{mx}g(x)$: ESA Jul-23 3c, May-24 3c, CW P24.

NOTE $g = \frac{1}{x}$: $u = -x$, $v = \log x$. $g = \frac{1}{x^2}$: $u = -\log x$, $v = -\frac{1}{x}$.

ADDED FOR CLARITY ESA Jul-23 3c · ESA May-24 3c · CW U2 P24

E4.2 · Results: $\frac{e^{mx}}{x}$ and $\frac{e^{mx}}{x^2}$ ESA ★★★

$$(D-m)^2y = \frac{e^{mx}}{x}: \quad y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$$

$$(D-m)^2y = \frac{e^{mx}}{x^2}: \quad y_p = -e^{mx}\log x$$

GIVES The ESA answers with $m = -3$ (Jul-23, Jun-24) and $m = 1$ (May-24); CW P24 with $m = -1$.

APPLIES After absorbing CF terms.

NOTE Raw $uy_1 + vy_2$ has an extra $-xe^{mx}$ (resp. $-e^{mx}$): a CF term.

ESA Jul-23 3c · ESA Jun-24 3a · ESA May-24 3c · CW U2 P24

B Symbol glossary

Symbol	Name	Meaning	Appears in
X	Right-hand side	Function of x on the right of the standard form $y'' + py' + qy = X$ (capital)	E1.1, E1.3
p, q	Coefficients	Constants of y', y after dividing by the coefficient of y''	E1.1
y_1, y_2	CF solutions	Linearly independent solutions of the homogeneous equation; $y_c = c_1 y_1 + c_2 y_2$	E1.1, E1.2
W	Wronskian	$y_1 y_2' - y_2 y_1'$; never zero here	E1.2, E3.1, E4.1
u, v	Varying parameters	$u = -\int \frac{y_2 X}{W} dx$, $v = \int \frac{y_1 X}{W} dx$; functions of x (some books: A, B)	E1.3
y_c, y_p	CF and PI	Complementary function and particular integral; $y = y_c + y_p$	E1.4
c_1, c_2	Arbitrary constants	In the CF only; absorb constant multiples of y_1, y_2	E1.4
$\alpha \pm i\beta$	Complex AE roots	$y_1 = e^{\alpha x}\cos\beta x$, $y_2 = e^{\alpha x}\sin\beta x$	E3.1
m	AE root	Repeated root in Topic 4 ($y_2 = xe^{mx}$)	E4.1, E4.2
$g(x)$	Reduced right-hand side	$X = e^{\alpha x} g$ or $e^{mx} g$; what is left after the exponentials cancel	E3.1, E4.1
z, D	Cauchy variable	$z = \log x$, $D = \frac{d}{dz}$ in W2.4; elsewhere $D = \frac{d}{dx}$ (clash)	W2.4
t	Substitution variable	$t = 1 + e^{-x}$ or $t = \cos x$ inside an integral (not time)	E2.1
$\log$	Natural logarithm	$\log_e$; strictly $\log\lvert\cdot\rvert$ in $\log\cos ax$, $\log\sin x$, $\log x$	E3.2, E4.2

Remark. Capital X is the right-hand side; small x the variable. u and v (the deck's names; some books write A, B) are the varying parameters, functions of x that replace the constants c_1, c_2. W is the Wronskian $W(y_1, y_2)$. In the Cauchy example the variable is $z = \log x$ and $D = \frac{d}{dz}$. $\log$ means $\log_e$; the CW answers write $\log\cos ax$, $\log\sin x$, $\log x$ without modulus bars — strictly $\log\lvert\cdot\rvert$, and the answers hold on any interval where the argument is positive.